

Emotion Overflow: an Interactive System to Represent Emotion with Fluid

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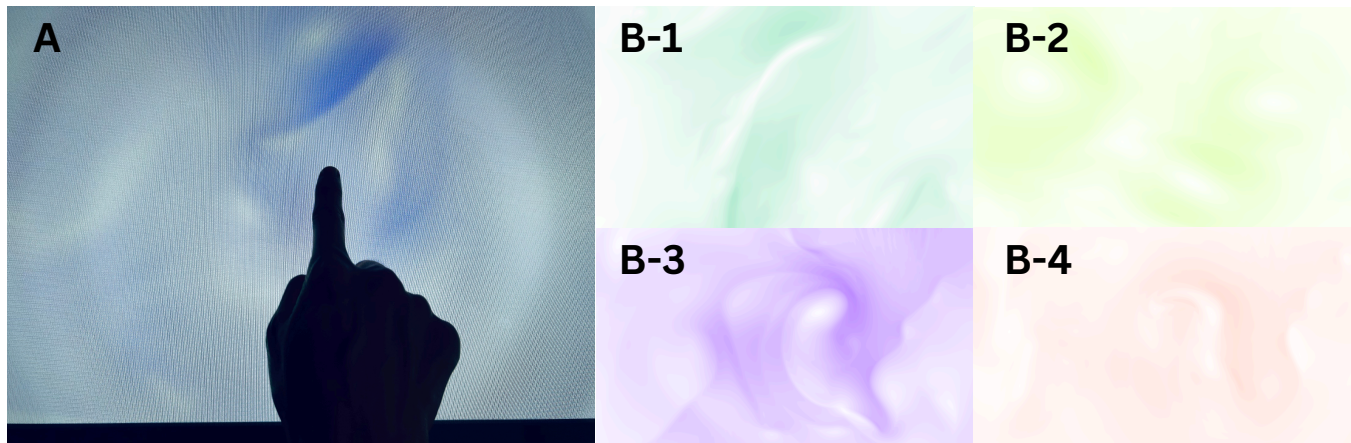


Figure 1: Overview of Emotion Overflow. A) Users are touching screen to interact with fluid. B) The representation of users emotion in fluid form. B-1 is satisfied, B-2 is joy, B-3 is sad, B-4 is nervous.

ABSTRACT

This paper presents Emotion Overflow, a novel system that transforms textual emotional input into interactive fluid visualizations. Utilizing a GPT model, our approach extracts emotion attributes from text and maps them onto fluid characteristics, creating a dynamic visual representation of emotional states. The system focuses on valence and arousal as key emotion attributes, translating them into fluid properties such as color, variance over time, and responsiveness. We demonstrate the application through two interactive experiences: a direct emotion visualization and an emotion guessing game. This work contributes to the field of emotional expression in human-computer interaction by offering a fluid, intuitive interface for exploring and communicating complex emotional states.

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1 INTRODUCTION

With the development of language model, it is becoming possible to analyze people's emotion from text. Recent Generative Pre-trained Transformer (GPT) predictions align significantly with human-provided appraisals and emotional labels [5]. While the extraction of emotions from text information has become sophisticated, the discussion around the representation systems has not been pursued. Emotion is often converted to emoji or color, but these are solid-state, and there are challenges in expressing emotions that are constantly in flux. Emotions are multifaceted, dynamic entities characterized by their complexity and gradations, often metaphorically likened to fluids in various languages, illustrating concepts such as “emotions overflowing” [1].

Therefore, we interpreted that representing emotion in fluid form, which contains color and fluid flow intuitively manipulated with touch interaction, aligns with human linguistic and cultural perceptions of emotion. Our contribution is mainly a proposal of new usage of emotional expression with computer graphics, following traditional expression styles. This work contributes to considering an interface between people and its emotion.

2 IMPLEMENTATION

Our approach utilizes a GPT model to translate emotion attributes into fluid characteristics for visualization. The process consists of two main stages: emotion attribute extraction and fluid characteristic generation. We focus on two key emotion attributes: First is valence, defined as the value associated with a stimulus, expressed on a continuum from pleasant to unpleasant or from attractive to

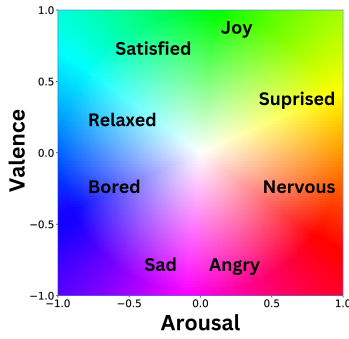


Figure 2: 2D Color Map of Valence and Arousal in our research. The fluid moves more quickly and dramatically as the Arousal increases, while a higher Valence causes the fluid to dissipate more rapidly.

aversive. The second is arousal, which is characterized as a state of excitement or energy expenditure linked to an emotion[3].

The GPT model analyzes the input text to determine values for these attributes based on the emotional content expressed. The emotion attributes extracted are subsequently mapped onto specific fluid characteristics to create a visual representation of the emotional state.

The color of the fluid is determined by a combination of both valence and arousal values as shown as Figure 2, with the color scheme designed to reflect the pleasantness and intensity of the emotion [2]. Variance over time, primarily influenced by the valence attribute, represents the fluid’s stability or volatility. This characteristic corresponds to the steadiness or fluctuation of the emotional state, providing a visual cue for the emotion’s constancy or variability. Lastly, fluid responsiveness is derived from the arousal attribute. This characteristic governs how quickly and dramatically the fluid reacts to user interactions, effectively mirroring the energy level associated with the emotion. Through this mapping process, the complex nuances of emotional states are translated into an interactive and visually dynamic fluid medium, allowing for an intuitive representation of the input emotion.

The Fluid in this experiment is coded using three.js, a JavaScript library for generating and presenting animated computer graphics within a web browser via WebGL. The implementation was inspired by the “fluid-three” GitHub repository, which implements Stable Fluids [4], a method that provides a foundation for simulating various kinds of fluids electronically.

3 DEMO APPLICATION

Emotion Overflow utilizes a text analysis module to interpret emotional content and a fluid dynamics engine to generate corresponding visual representations. Users interact with the system through a multi-touch screen interface, allowing for intuitive manipulation of the emotion-driven fluid simulations.

3.1 Direct Emotion Visualization

In the first demonstration, users experience the core functionality of Emotion Overflow. The user experience unfolds as follows: Users begin by entering a short text describing their current emotional



Figure 3: Users are sharing their emotion via fluids

state into a simple text box interface. Upon submission, the system analyzes the input to determine the predominant emotion. Based on this analysis, a unique fluid simulation is instantly generated on the multi-touch screen. Users can then interact freely with the fluid using various touch gestures. The fluid responds dynamically to each touch, providing an intuitive and engaging representation of the user’s emotional state.

3.2 Emotion Guessing Game

The second demonstration extends the concept into an interactive game that challenges users’ understanding of emotional expression through fluid dynamics. The user experience in this demo proceeds as follows: The system generates a fluid visualization based on a previous user’s emotional text input. New participants observe the fluid’s behavior and attempt to guess the emotion or emotional state that produced it. Users can interact with the fluid using the multi-touch screen to gain more insights into its characteristics. After guesses are made, the original input is revealed, leading to a group discussion. Participants analyze the accuracy of their guesses, evaluate the effectiveness of the fluid in conveying emotions, and consider the appropriate level of abstraction in emotional expression through fluid dynamics. The discussion focuses on finding the right balance between abstract representation and emotional clarity, contributing to the ongoing development of the Emotion Overflow system. Furthermore, interacting with the fluid forms of emotions left by others enriches communication, making it a more complex, multi-dimensional experience.

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